Part 12

Cell culture and its application

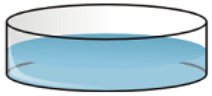


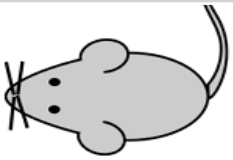




In vitro vs *In vivo*

Definition	
In vitro is a medical experiment or a study that is performed only in a laboratory dish or a test tube	*In vivo* describes a medical experiment or test that is performed on a living organism.





Cell culture is the process by which prokaryotic, eukaryotic cells are grown under controlled conditions/artificial environment



Sydney Ringer (1835-1910)

The 19th century, Sydney Ringer developed Ringer's solution suitable for maintaining the beating of an isolated animal heart outside of the body

In 1885 Roux removed a section of the medullary plate of an embryonic chicken and tamed it in a warm saline solution for 13 days

-> established the principle that tissues could live outside the body



Wilhelm Roux (1850 –1924)



The first to successfully grow artificial tissue culture (1907-1910)

Ross Granville Harrison (1870 –1959)

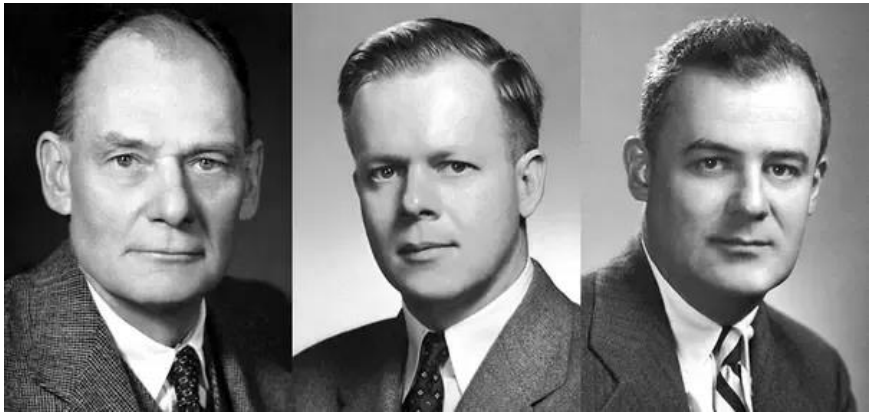
Polio and Nobel prizes





Paul Alexander in his iron lung.

In 1954, the Nobel Prize in Physiology or Medicine



John Enders, Thomas Weller, and Frederick Robbins

Discovery of a method of growing the virus in monkey kidney cell cultures

Animal cell culture

Two categories of cells



Primary cells

- Obtained directly form donor tissues

→ non-transformed, non immortalized cells directly from the tissue

- Only able to divide a finite number of times due to telomere loss

→ cell senescence



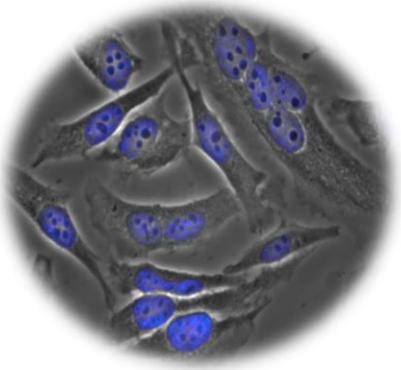
Established cell lines

- Able to escapes the normal constraints of the cell cycle, and grow indefinitely

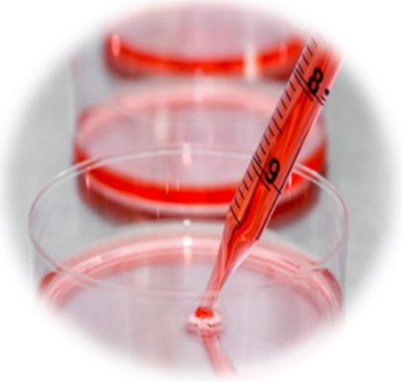
→ Extremely useful for long-term research

Obtained from: + Clinical tumors

+ Transforming primary cells with viral oncogenes or chemical treatments



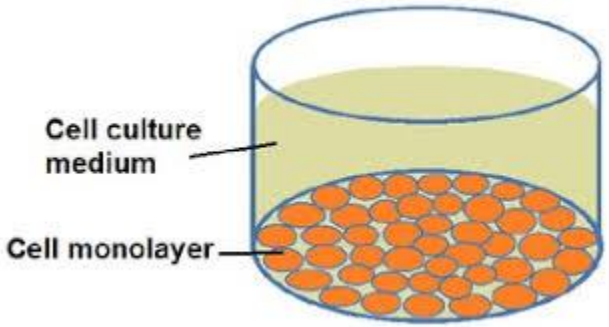
Able to growth indenfitely



Useful for long-term research

Hela Cell line

Cell growth environment



Adherent cultures



Suspension cultures

Appropriate culture media

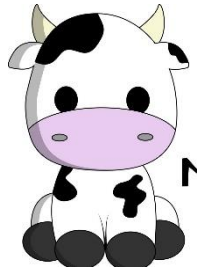


PH indicator

Sodium bicarbonate

Zwitterions (HEPES)

Common buffering systems



Keep the pH 7.2-7.4 with 5-10% CO_2





Vital supplements
(Macromolecules, lipids, growth factors)

Culture media

Fetal bovine serum (FBS)

Amino acids, inorganic salts, vitamins

Correct temperature



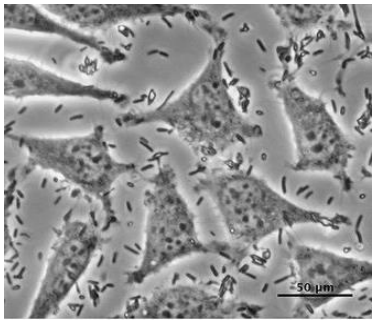


Mammalian cells ~ 37°C

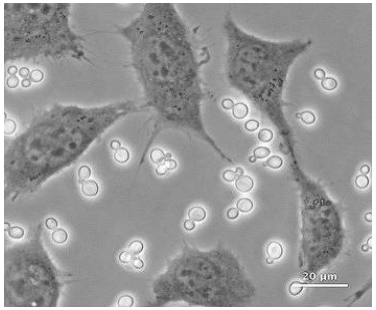


Cold -blooded animal

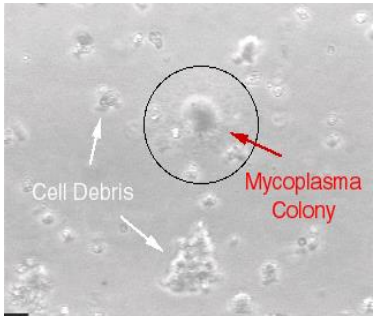
Sterilize environment



Bacteria



Fungi



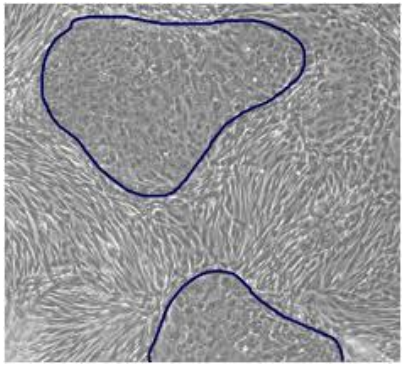
Mycoplasma



Laminar flow hood



Antibiotic

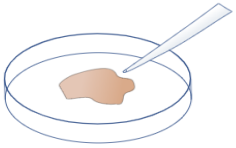


Cross contamination

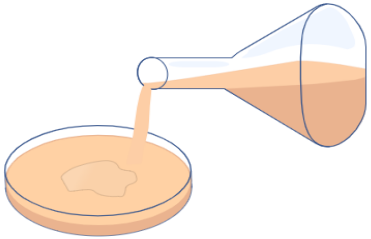
-> fault scientific conclusion

Bacterial culture

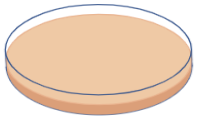
Pour plate method



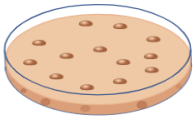
pipette inoculum onto sterile plate



add sterile medium

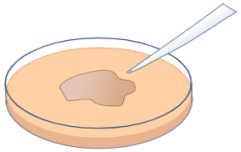


swirl to mix and incubate

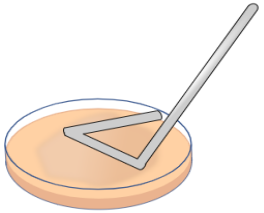


colonies grow
in and on medium

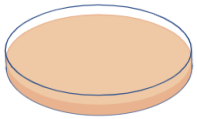
Spread plate method



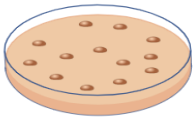
pipette inoculum onto the surface
of agar plate



spread evenly over the agar surface



incubate



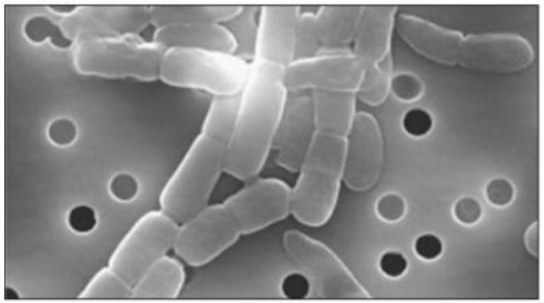
colonies grow only
on the surface of medium



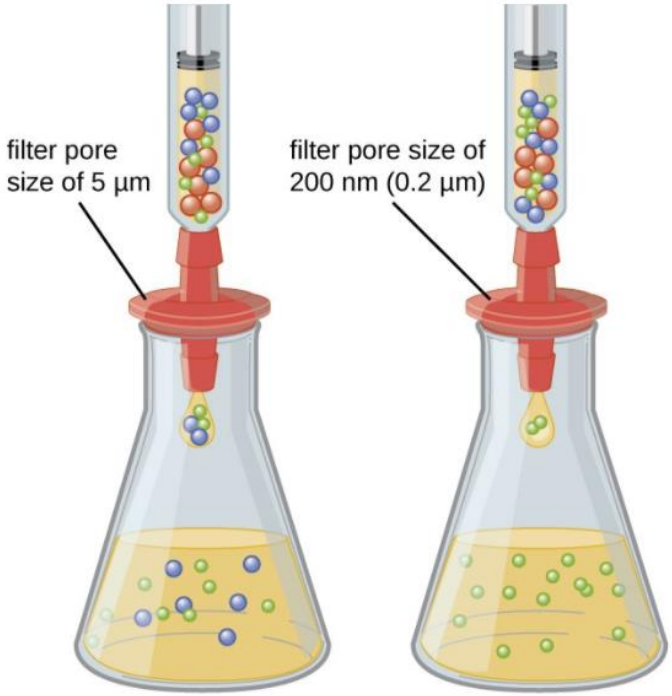
Culture of Viruses

Isolation of Viruses

Virions in the liquid medium can be separated from the host cells by either centrifugation or filtration



(a)

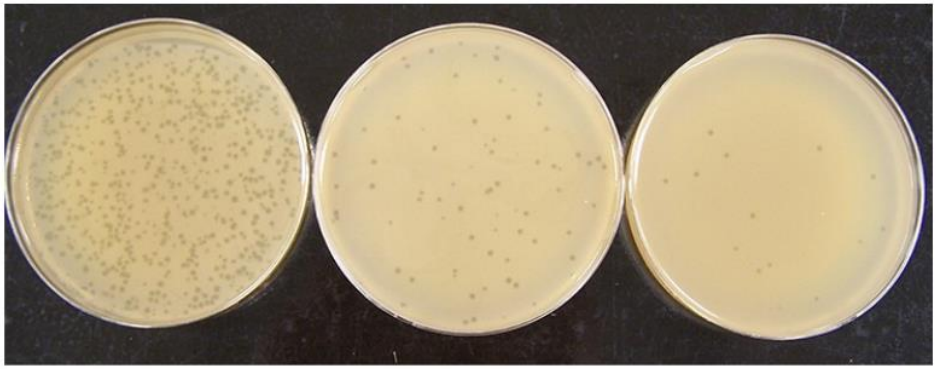


(b)

Culture of Viruses

- Identification and diagnosis of pathogenic viruses in clinical specimens,
- Production of vaccines
- Basic research studies





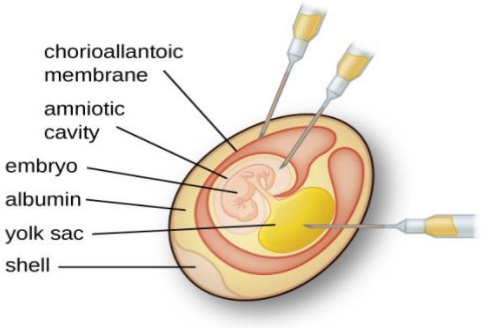
In vitro

Left: Flasks like this may be used to culture human or animal cells for viral culturing. Right: These plates contain bacteriophage T4 grown on an *Escherichia coli* lawn. Clear plaques are visible where host bacterial cells have been lysed. Viral titers increase on the plates to the left

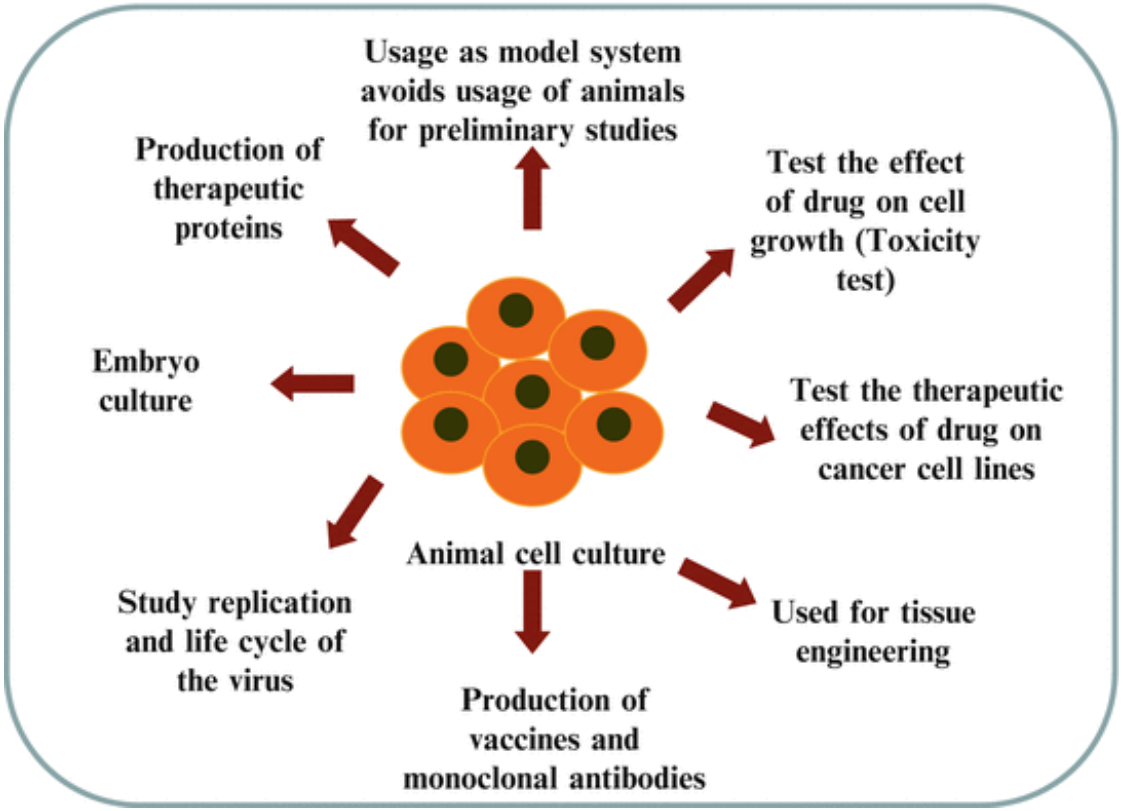
In vivo

Most of the influenza vaccine manufactured for annual flu vaccination programs is cultured in hens' eggs





The cells within chicken eggs are used to culture different types of viruses. Viruses can be replicated in various locations within the egg


Usage as model system avoids usage of animals for preliminary studies
Production of therapeutic proteins
Test the effect of drug on cell growth (Toxicity test)
Embryo culture
Test the therapeutic effects of drug on cancer cell lines
Animal cell culture
Study replication and life cycle of the virus
Used for tissue engineering
Production of vaccines and monoclonal antibodies

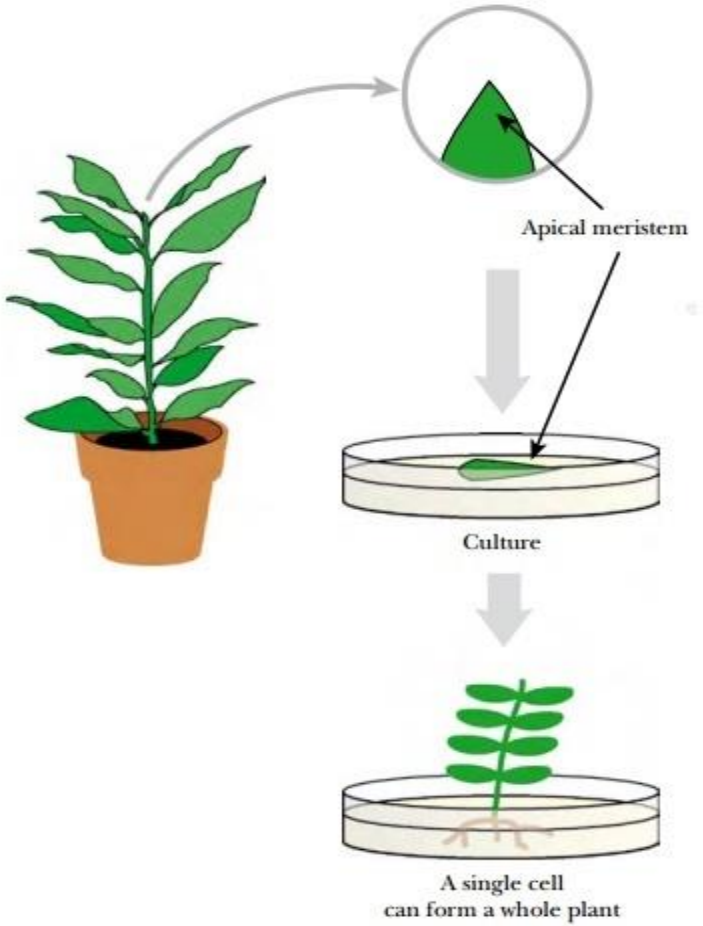
Plant tissue culture

Environmental conditions

- Nutrient medium
- Aseptic condition
- Aeration of the tissue

TOTIPOTENCY

PLASTICITY

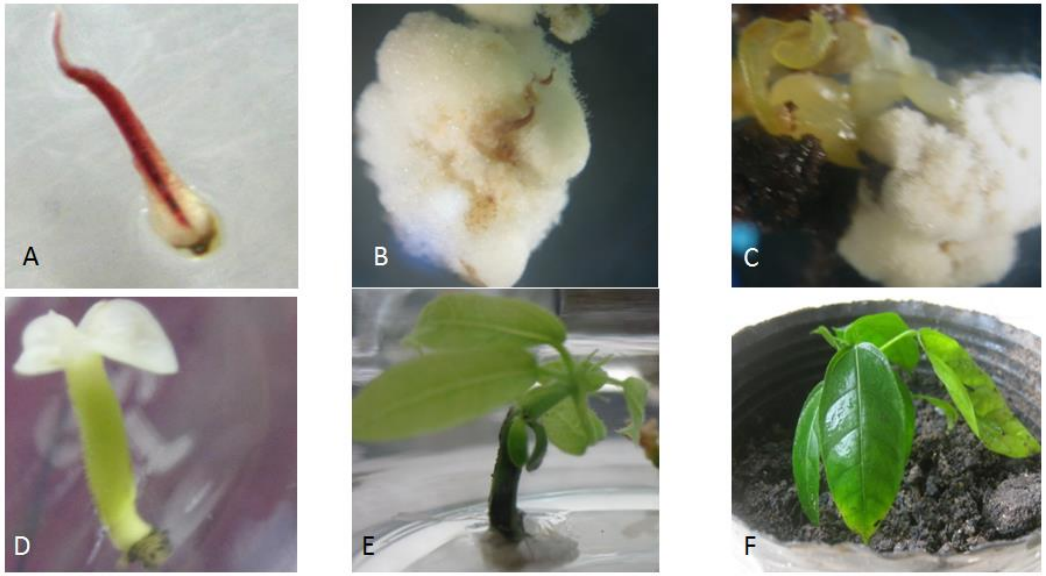


An entire plant can be regenerated from a single cell:
Small samples of tissue, or even single plant cells may be culture in *in vitro*. Under appropriate conditions, these may regenerate into complete plants

Plant tissue culture

Micro-propagation

- High rate
- Production of disease-free plants
- Cost-effective



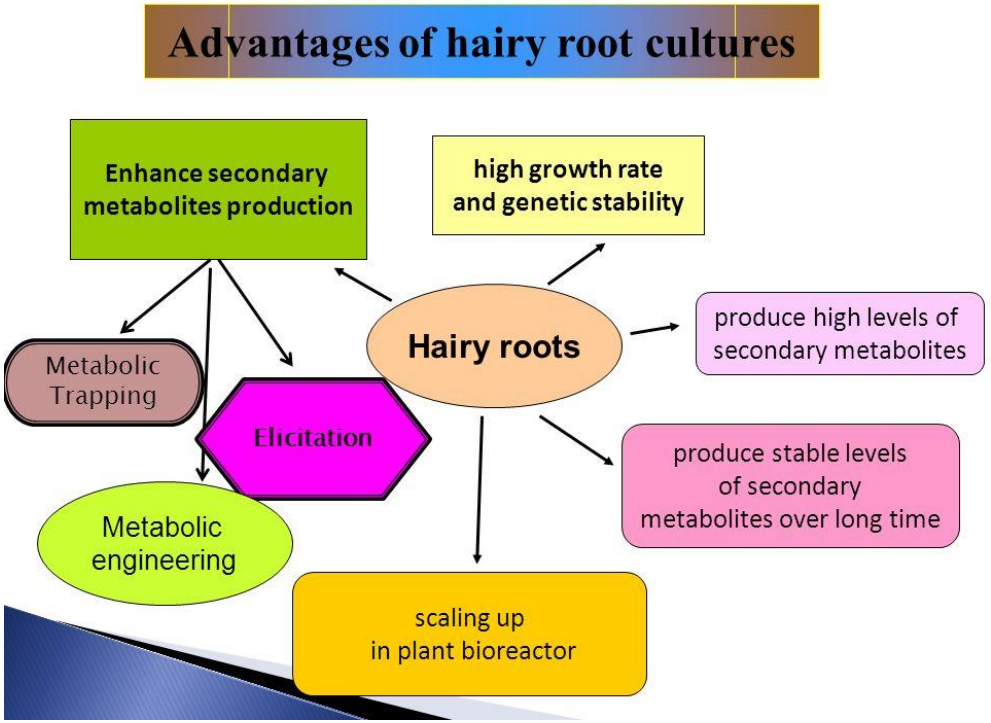
Somatic embryogenesis and plant regeneration from cultured staminodes explants of *Theobroma cacao* L.

Plant tissue culture

Secondary metabolite production

- Compounds- produced under controlled conditions
- Quality of product will be consistent
- Production of novel compounds
- Production time-less
- Labor costs- minimal





Plant tissue culture
GMO

Pros vs cons ?

